

Q7) Classification Metrics

Q7

	TP	FP	N	
P	45	15	FP	TP = 45
N	25	165	TN	TN = 165
	FN			FP = 15
				FN = 25

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\frac{45 + 165}{250} = 0.84$$

$$\text{Accuracy} = 84\%$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\frac{45}{45 + 15} = 0.75$$

$$\text{Precision} = 75\%$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\frac{45}{45 + 25} = 0.642$$

$$\text{Recall} = 64.2\%$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$2 \cdot \frac{0.75 \cdot 0.642}{0.75 + 0.642} = 2 \cdot \frac{0.4815}{1.392} = 0.69$$

$$F_1 = 0.69 = 69\%$$